Meridian Freight Website — Design Report

Document: Website Design Principles, Logic & Strategy **Prepared for:** Alexey Zaytsev, CEO **Site:** meridianexport.com
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1. Executive Summary

The Meridian Freight website is a production-grade, conversion-optimized corporate marketing platform for a machinery export and logistics company operating out of Albion, Iowa. It was rebuilt from scratch in March 2026, replacing an aging Create React App (CRA) single-page application with a modern Next.js 16 multi-page architecture.

Key design goals achieved:

Goal	Outcome
SEO visibility	From zero (SPA, invisible to Google) to **10/10 SEO score**
Lead generation	Email-gated freight calculator as **hero feature** — no competitor has this
Brand credibility	Real operational photography, verified statistics, outcome-focused copy
Performance	Static site generation (SSG) — sub-second loads, all green Core Web Vitals
Analytics	Full GA4 + Meta Pixel + CAPI + Google Ads tracking with consent mode
Mobile experience	Dedicated mobile bottom bar, responsive layouts, touch-optimized CTAs

The site serves a single business purpose: turn website visitors into qualified leads through two primary conversion paths — the contact form and the freight calculator — supported by WhatsApp as the preferred communication channel for international buyers.

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2. Strategic Context

2.1 Why the Rebuild

The previous CRA site had four fatal problems:

1. **Zero SEO** — Client-rendered SPAs are invisible to Google. Every competitor with server-rendered pages outranked Meridian.
2. **Dead framework** — CRA has no maintainer, no security patches, no modern tooling.
3. **No tracking** — The old site had zero analytics (no GA4, no Meta Pixel, no conversion tracking). $800+ in ad spend was wasted because the LP's pixel couldn't fire on the main domain.
4. **Narrow positioning** — The LP focused on JD parts for LATAM. The corporate site needed to present the full logistics offering.

2.2 Business Positioning

Meridian Freight is positioned as **the full-service machinery export specialist** — not just a shipping company, not just a packing company, but the single partner that handles the entire chain:

```
Equipment Pickup → Dismantling → Packing → Documentation → Shipping → Delivery
```

This "single partner" positioning is reflected in every design decision: the 6-service grid on the homepage, the 4-step process visualization, and the freight calculator that covers the complete cost chain.

2.3 Target Audience

Buyer Persona	Behavior	Design Impact
International farm/construction companies	Price-conscious, need transparent costs	Public pricing table + interactive calculator
Equipment dealers & resellers	Compare multiple logistics providers	Service detail pages with SEO-optimized keywords
First-time exporters	Anxious about customs, damage, hidden fees	FAQ page, process steps, trust signals
Returning customers	Know the process, want quick contact	Persistent WhatsApp widget, phone in header

2.4 Competitive Differentiation

The freight calculator (`/pricing/calculator`) is the **strategic differentiator**. No competitor in the heavy machinery export space offers a public, interactive cost estimator. This creates:

1. **Lead magnet** — Every calculation captures a qualified email with equipment type + destination
2. **Self-service** — Reduces "how much does it cost?" support queries
3. **Data quality** — Calculator leads are the most qualified (they know exactly what they want to ship)
4. **SEO advantage** — Interactive tools earn more engagement and links than static pages
5. **Conversion path** — Calculator → Detailed Quote → WhatsApp → Deal

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3. Brand Identity

3.1 Brand Attributes

Attribute	Expression
Professional	Clean layouts, Geist Sans typography, consistent spacing, no decorative clutter

Trustworthy	Verified statistics (500+ exports, 13 years), real photography, transparent pricing
Capable	6 distinct services, 27+ destination countries, multi-carrier partnerships
Accessible	WhatsApp as primary channel (international buyers prefer it), multilingual support (EN/RU/ES/AR)
Efficient	Process steps showing clear workflow, calculator delivering instant estimates

3.2 Brand Voice

The copy follows an **outcome-focused** approach rather than process description:

Instead of	We say
"We disassemble, pack, and ship"	"Ship $500K equipment without damage or customs delays"
"Professional services"	"500+ exports completed. Zero equipment damage claims."
"Contact us for pricing"	"Calculate your export cost in 60 seconds"

3.3 Visual Identity

- **Logo:** Meridian Freight wordmark with globe element (white on dark backgrounds, dark on light)
- **Primary brand color:** Sky-500 (#0EA5E9) — the "horizon blue" representing global ocean shipping
- **Photography:** Real operational photos (flat racks at ports, crane lifts, container loading) — never stock photos
- **Icons:** Lucide React — consistent line-weight icon system throughout

3.4 Contact Information Governance

All contact information lives in a single source of truth file (`lib/constants.ts`). Components **must** import from there — never hardcode phone numbers, emails, or URLs. This was a critical lesson: the old CRA site hardcoded a different phone number (+1-786-397-3888) in 6 components; the correct number is +1 (641) 516-1616.

```
COMPANY    → name, legalName, tagline, description, foundedYear
CONTACT    → phone, email, whatsappUrl, address, hours
SOCIAL     → facebook, instagram, youtube
SITE       → url, domain, ogImage
STATS      → projectsCompleted (500), yearsExperience (auto-calculated)
WAREHOUSES → Iowa HQ + 6 partner locations (CA, GA, IL, ND, TX, AB)
NAV_ITEMS  → 8 main items + 6 service sub-items
```

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4. Design Principles

4.1 Core Principles

The design system is called **"Meridian Horizon"** — subtitled **"Depth Without Lines."** Five principles guide every design decision:

1. Substance Over Style

> *A B2B buyer spending $50K+ on equipment shipping doesn't need entertainment — they need confidence. Every element must earn its screen space by communicating value, building trust, or enabling action.*

In practice: No decorative particles, no floating backgrounds, no typewriter effects. The anti-slop list (documented in the micro-interactions spec) explicitly bans 10 common AI-generated visual gimmicks.

2. Weight and Precision

> *Meridian moves heavy machinery across oceans. Animations should feel weighted — elements settle like a container on a dock, not bounce like a UI toy.*

In practice: All motion uses `ease-decelerate` (arriving/settling), never spring physics. Slide distances are modest (20px), durations are calibrated (500ms entrance, 150ms hover).

3. Progressive Disclosure

> *Don't overwhelm. Reveal information as the user scrolls, clicks, or advances through the calculator wizard.*

In practice: Homepage uses scroll-triggered reveals. FAQ uses accordion expansion. Calculator uses multi-step wizard. Contact info is split across header (phone), footer (full details), and dedicated contact page (map + form).

4. Single Path to Action

> *Every page should have one clear next step. Dual CTAs are acceptable (primary + secondary), but the primary action should be visually dominant.*

In practice: Primary buttons are solid blue (#0EA5E9). Secondary buttons are outlined. WhatsApp buttons are emerald green (#059669) — a separate visual track for the "talk to a human" escape hatch.

5. Mobile-First, Not Mobile-Only

> *60%+ of traffic is mobile (international buyers using WhatsApp links from ads). But desktop visitors (dealers, procurement teams) need full data tables and multi-step forms.*

In practice: Mobile gets a sticky bottom bar with 3 instant actions (WhatsApp/Phone/Quote). Desktop gets full navigation, side-by-side layouts, and the complete pricing table.

4.2 What We Do NOT Do

Explicitly banned patterns (from the Micro-Interactions Spec):

- Floating particles or abstract geometric shapes
- Gradient border glow animations
- Typewriter text effects
- Blob/morph backgrounds
- Infinite marquee scrollers
- Cursor trail effects
- Parallax on every section
- 3D card tilts on hover
- Bouncy spring physics
- Glassmorphism/frosted cards on content areas

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5. Information Architecture

5.1 Site Map

```
meridianexport.com/
│
├── /                          Homepage (long-scroll conversion page)
│
├── /services                  Services overview grid
│   ├── /machinery-packing     Dismantling & container packing
```

```
│   ├── /container-loading      Loading & export shipping
│   ├── /agricultural           Agricultural equipment specialization
│   ├── /equipment-sales        Sourcing & procurement
│   ├── /documentation          Export documentation & compliance
│   └── /warehousing            Storage & warehousing
│
├── /projects                   Portfolio gallery (8+ case studies)
│
├── /destinations               Destination country pages (27+ countries)
│   └── /[slug]                 Individual country detail
│
├── /equipment                  Equipment type pages (60+ types)
│   └── /[slug]                 Individual equipment detail
│
├── /pricing                    Static pricing table (60+ equipment types)
│   └── /calculator             Freight cost calculator (email-gated lead magnet)
│
├── /blog                       Industry articles
│   └── /[slug]                 Individual blog post
│
├── /about                      Company story, warehouses, differentiators
├── /faq                        20 categorized questions (JSON-LD FAQPage)
├── /contact                    Contact form + info cards
├── /privacy                    Privacy policy
└── /terms                      Terms of service
```

5.2 Why Multi-Page (Not SPA)

The decision to use individual routes for each service was critical for SEO. Each service page targets specific long-tail keywords:

Page	Target Keyword
`/services/machinery-packing`	"machinery container packing services"
`/services/container-loading`	"heavy equipment container loading"
`/services/agricultural`	"agricultural equipment export services"
`/services/equipment-sales`	"used farm equipment for export"
`/services/documentation`	"export documentation services USA"
`/services/warehousing`	"equipment storage warehousing USA"

A single-page SPA can never rank for these individual queries. The multi-page architecture also enables unique `<title>`, `<meta description>`, and JSON-LD structured data per page.

5.3 Navigation Architecture

Desktop: Horizontal nav with Services dropdown (6 sub-items) + "Get a Quote" CTA button **Mobile:** Hamburger → Sheet drawer with all nav items

Navigation items: Home → About → Services (dropdown) → Projects → Destinations → Pricing → FAQ → Blog → Contact

5.4 Cross-Linking Strategy

Internal linking is designed for SEO juice flow:

- Each service page links to 2-3 related services via `relatedSlugs`
- Service pages include service-specific FAQ entries
- Projects link to their associated service type
- Homepage sections link to their dedicated pages
- FAQ answers link to relevant service pages
- Equipment and destination pages cross-reference each other

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6. Visual Design System

6.1 Layout Grid

Token	Value	Usage
Max content width	`max-w-7xl` (80rem / 1280px)	All page content
Section padding (vertical)	`py-16 md:py-24` (64px → 96px)	Between major sections
Container padding (horizontal)	`px-4 sm:px-6 lg:px-8`	Standard content padding
Card padding	`p-6 sm:p-8`	Card internal spacing
Grid gap	`gap-6`	Consistent across all grids

6.2 Grid Patterns

Pattern	Breakpoints	Used For
Single column	`grid-cols-1` (mobile)	All sections on mobile
Two column	`sm:grid-cols-2`	Contact (form + info), about page cards
Three column	`lg:grid-cols-3`	Services grid, project grid, equipment badges
Sidebar layout	`lg:grid-cols-[1fr_400px]`	Calculator (wizard + estimate card)

6.3 Elevation & Depth

The "Depth Without Lines" philosophy means cards and sections create hierarchy through **subtle surface changes** rather than heavy borders:

Level	Treatment	Usage
L0 (base)	`bg-background` (white)	Page background
L1 (surface)	`bg-muted` (slate-50)	Alternating sections
L2 (card)	`bg-card` + `ghost-border` (1px @ 15% opacity)	Content cards
L3 (elevated)	`shadow-lg` + hover lift	Interactive cards on hover
L4 (overlay)	`bg-black/15 backdrop-blur-xl`	Modals, drawers

Ghost borders: A custom utility class (`ghost-border`) renders a 1px outline at 15% opacity — visible enough to define boundaries, invisible enough to avoid visual noise.

Glass utilities: Three levels for translucent surfaces:

- `.glass` — `bg-white/70 backdrop-blur-xl` (light glass)

- `.glass-heavy` — `bg-white/80 backdrop-blur-xl` (header, popover)
- `.glass-overlay` — `bg-black/15 backdrop-blur-xl` (dark overlay)

6.4 Corner Radius

Token	Value	Usage
`--radius`	`0.625rem` (10px)	Base radius
`--radius-sm`	6px	Small badges, tags
`--radius-md`	8px	Inputs, small buttons
`--radius-lg`	10px	Cards, modals
`--radius-xl`	14px	Large cards, hero CTAs

6.5 Button Hierarchy

Type	Style	Usage
Primary	Solid sky-500, white text, `h-12 px-8`, shadow	Main CTA ("Get a Quote")
Secondary / Outline	Border only, no fill, `h-10 px-5`	Alternative actions ("Our Services")
WhatsApp	Solid emerald-600, white text, WhatsApp icon	"Chat on WhatsApp" — separate visual track
Ghost	No border, no fill, hover: subtle bg	Navigation links, minor actions
Destructive	Red variant	Not used (no destructive actions on marketing site)

6.6 Dark Sections

Certain sections use a dark gradient background to create visual rhythm and emphasis:

- **Calculator CTA** on homepage — `bg-gradient-to-br from-slate-900 via-slate-800 to-slate-900`
- **Bottom CTAs** on interior pages — same dark treatment
- **Trust bar** — white text on dark, creating contrast against the hero

Text in dark sections uses `text-white` for headings, `text-sky-300` / `text-sky-400` for accent text, and `text-slate-300` for body copy.

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7. Typography

7.1 Font Stack

Font	Variable	Usage
Geist Sans	`--font-geist-sans`	All UI text — headings, body, navigation, buttons
Geist Mono	`--font-geist-mono`	Statistics, pricing numbers, technical data, code

Both fonts are loaded via `next/font/google` — zero layout shift, self-hosted, optimized for performance.

7.2 Type Scale

Element	Mobile	Desktop	Weight	Tracking

H1 (page hero)	`text-4xl` (36px)	`text-5xl` / `text-6xl` (48-60px)	Bold (700)	`tracking-tight`
H2 (section)	`text-3xl` (30px)	`text-4xl` / `text-5xl` (36-48px)	Bold (700)	`tracking-tight`
H3 (card title)	`text-lg` (18px)	`text-xl` (20px)	Semibold (600)	Normal
Body (primary)	`text-base` (16px)	`text-lg` (18px)	Regular (400)	Normal
Body (secondary)	`text-sm` (14px)	`text-base` (16px)	Regular (400)	Normal
Caption / meta	`text-xs` (12px)	`text-sm` (14px)	Medium (500)	`tracking-wide`
Stat number	`text-4xl` (36px)	`text-5xl` (48px)	Bold (700), `font-mono`	`tracking-tight`

7.3 Line Height

- Headings: `leading-tight` (1.25) — compact, impactful
- Body copy: `leading-relaxed` (1.625) — readable prose
- UI elements: `leading-normal` (1.5) — buttons, labels

7.4 Typography Rules

1. **Geist Mono for numbers:** All statistics, pricing figures, ZIP codes, transit times, and weights use `font-mono` to ensure tabular alignment and visual distinction from prose.
2. **No all-caps headings:** Headings use standard capitalization. All-caps is reserved for tiny eyebrow labels above section headings (e.g., "OUR SERVICES").
3. **Max readable width:** Prose blocks constrained to `max-w-2xl` or `max-w-3xl` for comfortable reading length.

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8. Color System

8.1 Design System Name: "Meridian Horizon"

The palette uses **oklch color space** (perceptually uniform) defined as CSS custom properties in `globals.css`:

8.2 Core Palette

Role	Color	oklch Value	Hex Reference	Usage
Primary	Sky-500	`oklch(0.685 0.169 222.7)`	#0EA5E9	Brand identity, CTAs, links, focus rings
Secondary (Action)	Emerald-500	`oklch(0.696 0.17 162.48)`	#10B981	WhatsApp buttons, success states
Tertiary (Depth)	Slate-800	`oklch(0.208 0.042 265.755)`	#1E293B	Dark sections, text anchor, navy depth
Surface	Slate-50	`oklch(0.984 0.003 247.858)`	#F8FAFC	Alternating section backgrounds
Background	White	`oklch(1 0 0)`	#FFFFFF	Main page background

8.3 Semantic Colors

Token	Value	Usage
`--foreground`	Slate-800	Primary text color
`--muted-foreground`	Slate-500	Secondary/descriptive text
`--border`	Slate-200	Card borders, dividers
`--ring`	Sky-500	Focus ring (matches primary)
`--destructive`	Red	Error states (form validation)

8.4 Chart Colors (Data Visualization)

```
chart-1: Sky-500     (primary blue)
chart-2: Emerald-500 (action green)
chart-3: Deep Sky    (darker blue)
chart-4: Navy        (deep blue)
chart-5: Slate-800   (anchor navy)
```

8.5 Color Usage Rules

1. **No dark mode.** This is a corporate marketing site — light theme only. The `.dark` block in `globals.css` exists for shadcn/ui compatibility but is never activated.
2. **Primary color for action.** Sky-500 is reserved for interactive elements: buttons, links, active states, focus rings. Never used as a background fill for large areas.
3. **Emerald is WhatsApp only.** The emerald-600 (#059669) channel is exclusively for WhatsApp CTAs. This creates a distinct visual association — "green button = talk to a human."
4. **Dark sections use slate-900/800/950.** Dark gradient sections create visual rhythm. Text in dark sections: `text-white` (headings), `text-sky-300` (accents), `text-slate-300` (body).
5. **Hover states darken, never lighten.** `hover:bg-primary/90` for buttons, `hover:text-sky-700` for links.

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9. Motion & Animation

9.1 Motion Philosophy

The motion system reflects the business: **weighted, directional, precise, purposeful, restrained.**

- **Weighted** — Elements feel like they have mass. Ease-out with deceleration, like a container settling on a dock.
- **Directional** — Motion implies movement along routes: left→right for progress, bottom→up for arrival.
- **Precise** — Clean, mechanical timing. No wobble, no overshoot. Like a calibrated crane.
- **Purposeful** — Every animation communicates state: arrival, change, relationship, or feedback.
- **Restrained** — If you notice the animation, it's too much. B2B buyers want confidence, not entertainment.

9.2 Motion Tokens

All animations share these constants (defined in `lib/motion.ts` for JS, `globals.css` for CSS):

Token	Value	Usage
`instant`	75ms	Button press, toggle

`fast`	150ms	Hover states, focus rings
`normal`	300ms	Card transitions, dropdowns
`entrance`	500ms	Scroll reveals, section entrance
`slow`	800ms	Counter animations, timeline draws

Easing curves:

Curve	Bezier	Usage
`decelerate`	`(0.0, 0.0, 0.15, 1.0)`	Entering elements (arriving, settling)
`accelerate`	`(0.4, 0.0, 1.0, 1.0)`	Exiting elements (departing)
`default`	`(0.25, 0.1, 0.25, 1.0)`	General purpose
`emphasis`	`(0.16, 1.0, 0.3, 1.0)`	Attention-drawing
`interactive`	`(0.25, 1.0, 0.5, 1.0)`	User-initiated actions

Stagger delays:

Context	Delay per item
Grid items	60ms
List items	40ms
Sections	120ms

9.3 Animation Inventory

Animation	Where	Trigger	Details
Scroll reveal (slide-up)	All major sections	Viewport entry (once)	`y: 20 → 0` , 500ms, decelerate ease
Scroll reveal (fade)	Images, videos	Viewport entry	Opacity only, 500ms
Scroll reveal (scale)	Accent sections	Viewport entry	`scale: 0.97 → 1` , 500ms
Stagger grid	Services, projects	Parent in view	60ms delay per child
Hero cascade	Homepage hero	Page load	0s, 0.15s, 0.3s, 0.5s, 0.6s delays
CountUp	Trust bar stats	In view	Numbers count from 0 to final value
Process line	Process steps (desktop)	Scroll-linked	Horizontal line draws as user scrolls
Button press	All buttons	Active state	`scale(0.98)` + shadow reduction, 75ms
Button hover	All buttons	Hover	`translateY(-1px)` + shadow-md, 150ms
Link underline	Text links	Hover	Underline slides in from left, 300ms
Card lift	Service/project cards	Hover	`-translate-y-1 + shadow-lg` , 150ms
Dropdown	Nav services menu	Click/hover	`scaleY(0.97 → 1)` , 300ms
Sheet slide	Mobile nav	Toggle	Slide from right, 300ms
Pulsing rings	Video play button	Continuous	2 concentric rings pulsing outward, 3s cycle
Shadow breathe	WhatsApp widget	Continuous	Subtle shadow pulse, 5s cycle
Nudge right	Arrow icons on CTAs	Continuous	4px horizontal nudge, 3s cycle

Shake	Form error	Validation fail	3px horizontal shake, 300ms
Draw checkmark	Form success	Submit success	SVG path draw, animated stroke-dashoffset

9.4 Accessibility: Reduced Motion

All animations respect `prefers-reduced-motion`:

```
@media (prefers-reduced-motion: reduce) {
  *, *::before, *::after {
    animation-duration: 0.01ms !important;
    animation-iteration-count: 1 !important;
    transition-duration: 0.01ms !important;
    scroll-behavior: auto !important;
  }
}
```

JS-side: the `ScrollReveal` component checks `useInView` with `once: true` and uses instant opacity transitions when reduced motion is active.

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10. Component Architecture

10.1 Component Hierarchy

```
Root Layout (layout.tsx)
├── ScrollProgress              Thin progress bar at top of page
├── Header                      Fixed sticky nav (desktop dropdown + mobile Sheet)
├── [Page Content]              Route-specific content
├── Footer                      6-column responsive grid
├── WhatsAppWidget              Floating emerald button → expandable chat popup
├── MobileBottomBar             Mobile-only sticky 3-button bar
(WhatsApp/Phone/Quote)
├── CookieConsent               Consent banner (localStorage-backed)
├── AttributionCapture          UTM/click ID capture (runs once)
├── GoogleAnalytics             GA4 + Consent Mode v2
├── MetaPixel                   Meta Pixel + consent revoke/grant
├── Vercel Analytics            Automatic page views
└── Vercel SpeedInsights        Core Web Vitals
```

10.2 Component Inventory (23 Custom Components)

Layout & Navigation (4)

Component	Type	Purpose
`header.tsx`	Client	Fixed sticky nav, desktop dropdown, mobile Sheet, WhatsApp CTA
`footer.tsx`	Server	6-column grid: services, company, resources, legal, contact, social

`mobile-bottom-bar.tsx`	Client	Sticky 3-button bar; hides on scroll down, shows on scroll up
`breadcrumbs.tsx`	Server	Semantic breadcrumb navigation with JSON-LD

Hero & Trust (2)

Component	Type	Purpose
`hero.tsx`	Server	Full-width hero with staggered cascade animation, 2 CTA buttons
`trust-bar.tsx`	Client	6 statistics with CountUp animation (intersection observer)

Content Sections (6)

Component	Type	Purpose
`services-grid.tsx`	Server	3-column grid of 6 services with StaggerItem hover lift
`process-steps.tsx`	Client	4-step timeline (vertical mobile, horizontal desktop with scroll-linked line)
`project-grid.tsx`	Server	3-column project portfolio with image + 4 metadata tags
`pricing-table.tsx`	Client	Searchable/filterable equipment pricing table
`video-section.tsx`	Client	YouTube lite-embed (lazy-loaded on play), custom pulsing play button
`faq-accordion.tsx`	Client	Expandable FAQ with GA4 tracking on each expansion

Forms & Interaction (3)

Component	Type	Purpose
`contact-form.tsx`	Client	6-field form + honeypot, Zod validation, success animation
`contact-info.tsx`	Server	4 contact cards (WhatsApp/Email/Address/Hours) + social icons
`freight-calculator/`	Client	Multi-step wizard: equipment → size → container → ZIP → country → email gate

Animation & Utility (4)

Component	Type	Purpose
`scroll-reveal.tsx`	Client	ScrollReveal wrapper (5 variants) + StaggerItem for grids
`scroll-progress.tsx`	Client	Optional scroll progress indicator
`tracked-contact-link.tsx`	Client	`<a>` wrapper for Server Components; fires GA4 + Pixel on click
`attribution-capture.tsx`	Client	UTM/click ID capture on mount

Analytics & Consent (4)

Component	Type	Purpose
`google-analytics.tsx`	Client	GA4 gtag init + Consent Mode v2
`meta-pixel.tsx`	Client	Meta Pixel init + consent revoke/grant
`cookie-consent.tsx`	Client	Banner + localStorage toggle
`whatsapp-widget.tsx`	Client	Floating button → expandable popup with session ref code

10.3 shadcn/ui Components (19 Installed)

```
accordion     alert-dialog   badge          button        card
dialog        dropdown-menu  input          label         navigation-menu
scroll-area   select         separator      sheet         skeleton
table         tabs           textarea       tooltip
```

All live in `components/ui/` as source code (not black-box packages). Styled with Radix primitives + Tailwind, using the `base-nova` variant with `neutral` base color.

10.4 Server vs Client Component Strategy

Default: Server Components. Only add `'use client'` when the component needs:

- Event handlers (onClick, onSubmit, onChange)
- Browser APIs (IntersectionObserver, localStorage)
- React hooks (useState, useEffect, useInView)
- Animation libraries (motion/react)

Client boundary pushed as low as possible. For example, `footer.tsx` is a Server Component but uses `TrackedContactLink` (a thin client wrapper) just for the click tracking on contact links.

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11. Content Strategy

11.1 Content Data Architecture

All content lives in typed TypeScript files in the `/content/` directory — not a CMS, not a database, not markdown files. This was a deliberate choice: content changes infrequently (equipment types, routes, FAQs are stable for months), and TypeScript provides compile-time safety.

File	Records	Exports	Purpose
`services.ts`	6	`ServicePage[]` , helpers	Service detail pages with cross-links
`projects.ts`	8+	`Project[]`	Portfolio case studies with images
`pricing.ts`	40+	`EquipmentPricing[]` , `deliveryRates[]`	Static pricing reference table
`faq.ts`	20	`FaqEntry[]` , `homepageFaq`	Categorized FAQ (General, Shipping, Documentation, Pricing)
`equipment.ts`	60+	`EquipmentType[]`	Equipment type detail pages
`destinations.ts`	27+	`Destination[]`	Country detail pages
`blog.ts`	Multiple	`BlogPost[]`	Blog articles

11.2 Content Type Schemas

Each content type is fully typed with interfaces that enforce required fields:

```
// Service pages include cross-linking and SEO data
interface ServicePage {
  slug: string;           // URL-safe identifier
  title: string;          // Full page title
```

```
  shortTitle: string;     // Grid card display name
  description: string;    // Brief (1-2 sentences)
  longDescription: string;// Full service explanation
  icon: string;           // Lucide icon name
  keywords: string[];     // SEO keywords for metadata
  equipmentTypes: string[];// Equipment this service handles
  relatedSlugs: string[]; // Cross-links to related services
  faqs?: FaqEntry[];      // Service-specific FAQ entries
}
```

11.3 Copy Principles

Based on the SEO & Copywriting audit (2026-03-18):

1. **Outcome over process** — Headlines communicate what the buyer gets, not what Meridian does
2. **Specific over generic** — "500+ exports to 27 countries" beats "trusted worldwide"
3. **Fear-addressing** — B2B buyers fear equipment damage, customs delays, hidden fees, and lost parts. Copy addresses these directly.
4. **Social proof** — Verified metrics from operations (not fabricated testimonials)
5. **CTA clarity** — Every page ends with a clear next step

11.4 Image Strategy

Based on the Image Strategy spec (2026-03-18):

- **Hero:** Real operational photo (JD W260 on Hapag-Lloyd flat rack) — never stock
- **Projects:** Actual equipment photos from Meridian's operations showing carrier branding (Hapag-Lloyd, Maersk, COSCO)
- **Optimization:** All images via `next/image` (WebP conversion, lazy loading, responsive srcset)
- **Hero priority:** `priority` prop + `fetchPriority="high"` for LCP optimization

---

12. SEO Strategy

12.1 SEO Score: 10/10

The site achieves comprehensive SEO coverage across all dimensions:

12.2 Technical SEO

Feature	Implementation
Sitemap	Auto-generated via `app/sitemap.ts` — 40+ pages with priorities + change frequencies
Robots.txt	`app/robots.ts` — allows all, references sitemap
Canonical URLs	Absolute URLs on all pages via `pageMetadata()` helper
Meta descriptions	All pages, 120-160 character range
Security headers	HSTS, X-Frame-Options, X-Content-Type-Options, Referrer-Policy, Permissions-Policy
IndexNow	Key verification + bulk submit script for Bing/Yandex instant indexing
Search Console	Google + Bing verification files
llms.txt	Model training opt-out

Static generation	All pages SSG via `generateStaticParams()` — fastest possible TTFB

12.3 Structured Data (JSON-LD)

Schema Type	Pages	Rich Result
`LocalBusiness`	Homepage (layout)	Business panel in Google
`WebSite`	Homepage (layout)	Sitelinks search box
`FAQPage`	Homepage, FAQ, service pages	FAQ rich snippets
`Service`	Service detail pages	Service markup
`Article`	Blog posts	Article rich snippets
`ContactPage`	Contact page	Contact information
`WebApplication`	Calculator page	App markup
`AggregateOffer`	Pricing page	Price range display
`VideoObject`	Homepage (video section)	Video rich snippet
`BreadcrumbList`	All interior pages	Breadcrumb trail in SERP
`ItemList`	Projects page	List markup

12.4 On-Page SEO

Every page receives via `pageMetadata()` helper:

- Unique `<title>` (template: `%s | Meridian Export`)
- Unique `<meta description>` (120-160 characters)
- Keywords array (from service/equipment/destination data)
- Canonical URL (absolute)
- OpenGraph tags (title, description, 1200x630 image)
- Twitter Card tags (summary_large_image)

12.5 Content Grouping (GA4)

Pages are assigned to content groups for analytics:

```
Homepage, Services, Equipment, Destinations, Calculator,
Pricing, Projects, About, Contact, FAQ, Blog, Legal
```

---

13. Conversion & Lead Generation

13.1 Two Primary Conversion Paths

```
Path 1: Contact Form                    Path 2: Freight Calculator

┌──────────────────────┐                ┌──────────────────────────┐
│ Any page → Contact   │                │ Pricing → Calculator     │
│ Form submission      │                │ Multi-step wizard        │
│ → Qualified lead     │                │ → Email gate             │
```

```
| (name, email,       |                 | → Qualified lead         |
|  equipment, message)|                 | (email, equipment,       |
|                     |                 |  destination, container) |
| Value: $500 (GA4)   |                 | Value: $300 (GA4)        |
└────────────────────┘                 └──────────────────────────┘
```

13.2 Secondary Conversion: WhatsApp

WhatsApp is the **preferred channel for international buyers** (especially LATAM, Middle East, Central Asia). The site provides 15+ WhatsApp touchpoints:

Location	Format
Floating widget (all pages)	Expandable popup with pre-filled message + session ref code
Mobile bottom bar	Instant tap-to-chat button
Header CTA	Desktop nav button
Hero section	Primary/secondary CTA
Service page CTAs	Bottom-of-page CTA
Calculator results	"Discuss this estimate on WhatsApp"
Contact info cards	Contact page

Each WhatsApp click generates a unique ref code (MF-XXXXX) tracked via `lib/wa-attribution.ts` for conversion attribution.

13.3 Lead Pipeline (Server Actions)

Both the contact form and calculator use the same server-side pipeline:

```
1. Client-side Zod validation
2. Submit → Server Action
3. Server-side Zod re-validation
4. Honeypot check (silent success for bots — no error message)
5. Supabase INSERT → leads table (best-effort)
6. Resend email to owner (alex.z@meridianexport.com) — MUST succeed
7. Resend auto-reply to visitor (best-effort)
8. Slack notification → #meridian-email-intake (best-effort)
9. Meta CAPI Lead event with hashed PII (best-effort)
10. Return success + eventId → client fires GA4 + Pixel dedup event
```

13.4 Calculator as Lead Magnet

The freight calculator is architecturally the most sophisticated feature:

Data sources: Live rates from Supabase (`equipment_packing_rates` + `ocean_freight_rates` tables)

Calculation engine: `lib/freight-engine-v2.ts` — 69 unit tests with hand-verified formulas

Two container types with different routing logic:

- **40HC** — Equipment routed through Albion, IA → Chicago rail → ocean
- **Flatrack** — Equipment shipped directly to nearest US port (optimized across 4 ports)

Email gate strategy: Equipment selection and route visualization are free (Steps 1-5). The detailed cost breakdown is gated behind email capture (Step 6). This ensures the email lead includes complete context (equipment type, size, container, destination).

Integrity: Client-side preview is for UX only. On email submission, the server action re-fetches rates from Supabase and re-calculates independently — the client estimate is never trusted.

---

14. Analytics & Tracking

14.1 Stack

Layer	Tool	Purpose
Client-side web analytics	GA4 (G-W661JN5ED4)	Page views, events, conversions
Client-side ad tracking	Meta Pixel	PageView, Lead, Contact events
Server-side ad tracking	Meta CAPI	Server-side Lead event (deduped with Pixel)
Ad conversion	Google Ads tag	Conversion tracking for leads
Platform analytics	Vercel Analytics	Automatic page views
Performance	Vercel Speed Insights	Core Web Vitals

14.2 Consent Architecture

Both GA4 and Meta Pixel implement consent mode — scripts **always load**, but default to restricted/cookieless mode:

Tool	Default State	On Consent Accept	On Consent Decline
GA4	`analytics_storage: denied` (cookieless)	`analytics_storage: granted`	Stays denied
Meta Pixel	`fbq('consent', 'revoke')`	`fbq('consent', 'grant')`	Stays revoked
Meta CAPI	Always fires (server-side)	No change	No change

Critical rule: The Meta Pixel must **always** load (not be consent-gated at the script level). A previous implementation that only loaded the pixel after consent caused **$811 in wasted ad spend** with zero PageView tracking. The `fbq('consent', 'revoke')` pattern is the correct approach.

Consent stored in `localStorage['cookie-consent']` as `"accepted"` or `"declined"`.

14.3 Key Events (Conversions)

Event	Value	Trigger	Components
`generate_lead`	$500	Contact form submit	`contact-form.tsx`
`generate_lead`	$300	Calculator email submit	`calculator-wizard.tsx`
`contact_whatsapp`	—	WhatsApp link click	15+ locations
`contact_phone`	—	Phone link click	Header, footer, mobile bar
`contact_email`	—	Email link click	Footer, contact info

14.4 Micro-Conversion Events

Event	Component	Trigger

`calculator_start`	Calculator wizard	Equipment selected
`calculator_step`	Calculator wizard	Step advance
`calculator_complete`	Calculator wizard	Ready for email gate
`video_play`	Video section	Play button click
`faq_expand`	FAQ accordion	Item opened
`cta_click`	Various	CTA button click (with location param)

14.5 UTM Attribution

- **Storage:** 30-day first-party cookie (`mf_attribution`)
- **Fallback:** sessionStorage for backward compatibility
- **Captured:** `utm_source` , `utm_medium` , `utm_campaign` , `utm_term` , `utm_content` , `gclid` , `fbclid` , `msclkid`
- **Sent with:** Every form/calculator submission as metadata

---

15. Technical Architecture

15.1 Tech Stack

Layer	Technology	Version
Framework	Next.js	16 (App Router, React 19)
Language	TypeScript	5 (strict mode)
Styling	Tailwind CSS	4 (oklch, `@theme inline`)
Components	shadcn/ui	base-nova style, 19 components
Animation	motion (formerly framer-motion)	12.38
Validation	Zod	4.3
Email	Resend	6.9
Database	Supabase	REST API (shared with chatbot)
Icons	Lucide React	0.577
Testing	Vitest	4.1 (69 freight engine tests)

15.2 Rendering Strategy

All pages are statically generated (SSG) via `generateStaticParams()` . There are zero server-rendered-on-request pages. Server Actions (contact form, calculator) are serverless functions.

This means:

- Sub-second TTFB worldwide (served from CDN edge)
- Zero server cost for page views
- Build-time validation of all routes

15.3 Server Actions vs API Routes

The site uses **Server Actions** for all data mutations (contact form, calculator submission). API routes are only used for:

- WhatsApp click tracking (`/api/track/wa-click`)

- IndexNow verification (`/api/indexnow-verify`)

15.4 Environment Variables

Variable	Service	Required
`RESEND_API_KEY`	Email sending	Yes
`SUPABASE_URL`	Lead storage + freight rates	No (graceful fallback)
`SUPABASE_SERVICE_ROLE_KEY`	Database auth	No
`SLACK_BOT_TOKEN`	Slack notifications	No
`NEXT_PUBLIC_GA_MEASUREMENT_ID`	GA4	No
`NEXT_PUBLIC_META_PIXEL_ID`	Meta Pixel	No
`META_PIXEL_ID` + `META_ACCESS_TOKEN`	Meta CAPI	No
`NEXT_PUBLIC_GOOGLE_ADS_ID`	Google Ads	No
`INDEXNOW_KEY` + `INDEXNOW_SECRET`	IndexNow	No

Graceful degradation: If Supabase is not configured, the calculator shows "Calculator Unavailable" with contact CTAs. The static pricing table is unaffected.

15.5 Deployment

- **Platform:** Vercel (project: `meridian-freight-export`)
- **Domain:** meridianexport.com
- **Auto-deploy:** Push to `main` → automatic production deployment
- **Dependabot:** Auto-merge enabled via GitHub Actions
- **Security headers:** HSTS, X-Frame-Options, X-Content-Type-Options, Referrer-Policy, Permissions-Policy configured in `next.config.ts`

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16. Performance Strategy

16.1 Performance Targets

Metric	Target	Technique
LCP	< 2.5s	SSG, `priority` on hero image, `next/image` optimization
INP	< 100ms	Minimal client JS, lazy load non-critical components
CLS	< 0.1	`next/image` dimensions, `next/font` (zero layout shift)
TTI	< 3s	Static pages, code splitting
First Load JS	< 150KB	Tree shaking, dynamic imports
Lighthouse	90+ all categories	Semantic HTML, ARIA, performance optimization

16.2 Key Optimizations

Optimization	Implementation
Static generation	All pages SSG — served from CDN edge
Image optimization	`next/image` with WebP, lazy loading, responsive `sizes`

Font optimization	`next/font/google` (Geist) — self-hosted, zero CLS
Video lazy-load	YouTube iframe only loads on play button click (not on page load)
Code splitting	`'use client'` boundary pushed as low as possible
Bundle size	Tree-shaking via Turbopack, no unused shadcn components imported
CSS performance	Tailwind v4 with JIT — only used classes in output
Animation performance	GPU-accelerated transforms only (translate, scale, opacity)

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17. Accessibility

17.1 Standards

The site targets **WCAG 2.1 AA** compliance:

Area	Implementation
Semantic HTML	`<section>`, `<article>`, `<nav>`, `<main>`, `<header>`, `<footer>` landmarks
ARIA labels	All icon-only buttons, interactive elements, and navigation items labeled
Focus management	shadcn/ui Radix primitives provide built-in focus rings and keyboard navigation
Keyboard navigation	Full keyboard support for nav, accordion, forms, modals (Escape to close)
Color contrast	Slate-800 on white (text) meets 4.5:1 minimum ratio
Form labels	Explicit `<Label htmlFor>` on all form inputs
Reduced motion	Global `prefers-reduced-motion` override disables all animations
Skip links	Header includes keyboard-accessible skip navigation
Image alt text	All `next/image` components include descriptive `alt` attributes

---

18. Mobile Strategy

18.1 Responsive Breakpoints

Breakpoint	Width	Layout Changes
Mobile	< 640px	Single column, stacked sections, hamburger nav, mobile bottom bar
Tablet	640-1023px	2-column grids, collapsible nav
Desktop	1024px+	Full 3-column grids, horizontal nav, side-by-side layouts

18.2 Mobile-Specific Components

Component	Behavior
Mobile Bottom Bar	Sticky 3-button bar (WhatsApp / Phone / Get Quote). Hides on scroll down, shows on scroll up. Hidden on `lg:` and above.
Header (mobile)	Hamburger → Sheet drawer sliding from right with all nav items

WhatsApp Widget	Same on all sizes but positioned above mobile bottom bar
Footer	Stacks from 6 columns to single column on mobile

18.3 Touch Optimizations

- All tap targets minimum 44x44px (Apple HIG compliance)
- WhatsApp/Phone buttons are `h-12 px-8` (48px height)
- Bottom bar buttons span full width in 3-column grid
- Form inputs are `h-10` minimum with generous touch padding

Appendix A: File Structure

```
meridian-freight-website/
├── app/
│   ├── layout.tsx                  Root layout
│   ├── page.tsx                    Homepage (10 sections)
│   ├── globals.css                 Design tokens + animations
│   ├── sitemap.ts                  Auto-generated sitemap
│   ├── robots.ts                   robots.txt
│   ├── actions/                    Server Actions
│   ├── api/                        API routes (tracking)
│   ├── about/page.tsx
│   ├── contact/page.tsx
│   ├── faq/page.tsx
│   ├── pricing/page.tsx
│   ├── pricing/calculator/page.tsx
│   ├── projects/page.tsx
│   ├── services/[slug]/page.tsx
│   ├── equipment/[slug]/page.tsx
│   ├── destinations/[slug]/page.tsx
│   ├── blog/[slug]/page.tsx
│   ├── privacy/page.tsx
│   └── terms/page.tsx
├── components/
│   ├── ui/                         19 shadcn/ui primitives
│   ├── header.tsx
│   ├── footer.tsx
│   ├── hero.tsx
│   ├── trust-bar.tsx
│   ├── services-grid.tsx
│   ├── process-steps.tsx
│   ├── project-grid.tsx
│   ├── video-section.tsx
│   ├── faq-accordion.tsx
│   ├── contact-form.tsx
│   ├── contact-info.tsx
│   ├── scroll-reveal.tsx
│   ├── pricing-table.tsx
```

```
│   ├── freight-calculator/         Multi-step calculator wizard
│   ├── whatsapp-widget.tsx
│   ├── mobile-bottom-bar.tsx
│   ├── cookie-consent.tsx
│   ├── google-analytics.tsx
│   ├── meta-pixel.tsx
│   └── tracked-contact-link.tsx
├── content/
│   ├── services.ts                 6 service detail pages
│   ├── projects.ts                 Portfolio case studies
│   ├── pricing.ts                  Static pricing reference
│   ├── faq.ts                      20 FAQ entries
│   ├── equipment.ts                60+ equipment types
│   ├── destinations.ts             27+ countries
│   └── blog.ts                     Blog articles
├── lib/
│   ├── constants.ts                Single source of truth (contact, company, URLs)
│   ├── motion.ts                   Animation tokens
│   ├── tracking.ts                 GA4/Pixel/attribution helpers
│   ├── metadata.ts                 pageMetadata() helper
│   ├── schemas.ts                  Zod validation schemas
│   ├── freight-engine-v2.ts        Calculator formulas (69 tests)
│   ├── supabase-rates.ts           Rate fetching
│   ├── meta-capi.ts                Server-side Meta events
│   └── slack.ts                    Slack notifications
├── docs/
│   ├── WEBSITE-DESIGN-REPORT.md    ← This document
│   ├── GA4-DASHBOARD-SPEC.md       Analytics framework
│   ├── MICRO-INTERACTIONS-SPEC.md  Animation guidelines
│   └── plans/                      Design specs and PRDs
└── public/
    ├── og.jpg                      Open Graph image (1200x630)
    ├── images/                     Project photos
    └── logos/                      Company logos
```

Appendix B: Design Decision Log

Decision	Chosen	Rejected	Why
Framework	Next.js 16	CRA, Gatsby, Astro	App Router, SSG, Server Actions, Turbopack, Vercel-native
Styling	Tailwind CSS 4	CSS Modules, Styled Components	shadcn-native, oklch colors, JIT compilation
Components	shadcn/ui	Material UI, Chakra UI, custom	Source code ownership, Radix accessibility, customizable
Animation	motion	CSS-only, GSAP	React-native integration, `useInView`, stagger support
Font	Geist	Inter, DM Sans	Vercel default, professional, `next/font` optimized

Dark mode	Disabled	Enabled	Corporate marketing site — light theme only
CMS	None (TypeScript files)	Sanity, Contentful	Content changes infrequently, compile-time safety
Auth	None	Clerk, Auth0	Public marketing site — no user accounts
i18n	English only	Multi-language	Corporate site is English; localized pages are the LP's job
Database	Supabase (shared with chatbot)	Separate instance	Same rates, same leads table, unified data
Email	Resend	SendGrid, SES	Already verified on meridianexport.com domain
Pricing	Public + calculator	Gated behind form	Transparency builds trust; calculator captures qualified leads
Contact channel	WhatsApp primary	Email only	International buyers (LATAM, Central Asia, Middle East) prefer WhatsApp

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End of document.